

Lab Experiments 13 – 17

SET 13 – Q1

Create a Map Chart

Install package (Run only once)

```
install.packages("leaflet")
```

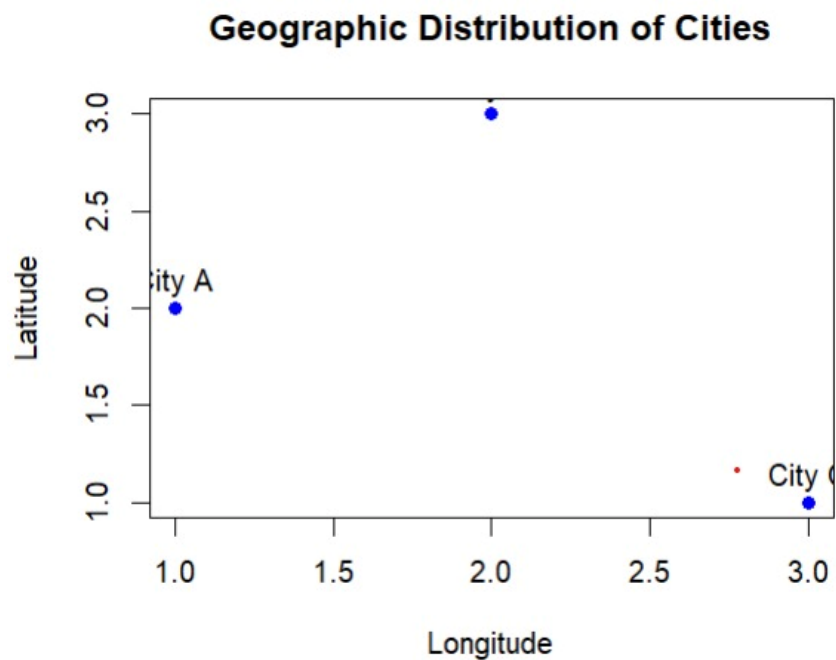
```
library(leaflet)
```

Dataset

```
geo <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Latitude = c(13.0827, 12.9716, 17.3850),  
  Longitude = c(80.2707, 77.5946, 78.4867),  
  Population = c(500000, 700000, 600000)  
)
```

Map Chart

```
leaflet(geo) %>%  
  addTiles() %>%  
  addMarkers(  
    lng = ~Longitude,  
    lat = ~Latitude,  
    popup = ~paste(  
      "<b>", City, "</b><br>",  
      "Population:", Population  
    )  
  )  
)
```



SET 13 – Q2

Scatter Plot (Population vs Temperature)

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

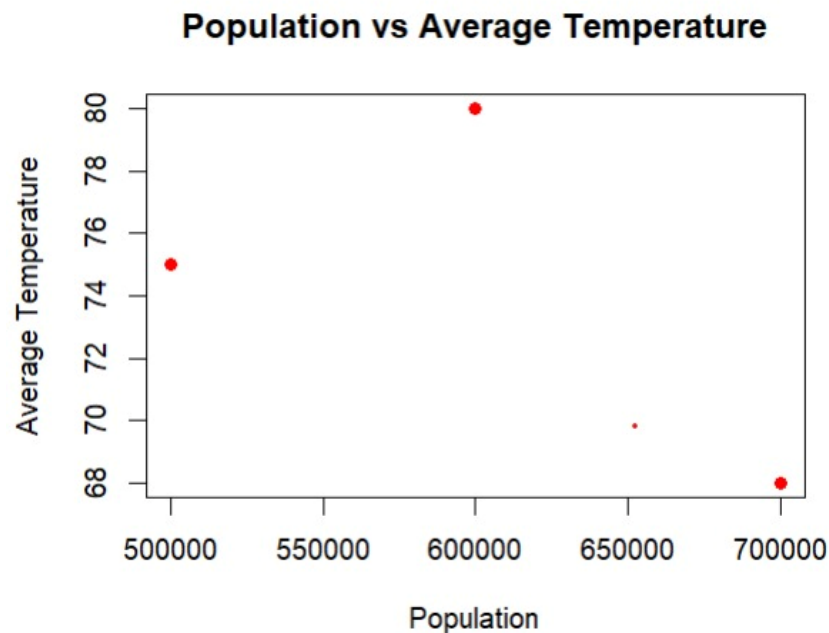
```
geo <- data.frame(  
  City=c("City A","City B","City C"),  
  Population=c(500000,700000,600000),  
  AvgTemperature=c(75,68,80)  
)
```

```
ggplot(geo,  
  aes(x=AvgTemperature,  
    y=Population,  
    label=City))+
```

```
geom_point(size=5,color="blue")+
```

```
geom_text(vjust=-1)+
```

```
labs(title="Population vs Average Temperature",  
      x="Average Temperature",  
      y="Population")
```



SET 13 – Q3

Display Geographic Data Table

```
geo <- data.frame(  
  City=c("City A","City B","City C"),  
  Population=c(500000,700000,600000),  
  AvgTemperature=c(75,68,80),  
  Elevation=c(1000,800,1200)  
)
```

```
cat("Geographic Data\n")  
print(geo)
```

Geographic Data

```
> print(geo)
```

	City	Population	AvgTemperature	Elevation
1	City A	5e+05	75	1000
2	City B	7e+05	68	800
3	City C	6e+05	80	1200

```
> |
```

SET 13 – Q4

Dashboard Equivalent in R

```
install.packages(c("leaflet","ggplot2","gridExtra"))
```

```
library(leaflet)
```

```
library(ggplot2)
```

```
library(gridExtra)
```

```
geo <- data.frame(  
  City=c("City A","City B","City C"),  
  Population=c(500000,700000,600000),  
  AvgTemperature=c(75,68,80),  
  Elevation=c(1000,800,1200),  
  Latitude=c(13.0827,12.9716,17.3850),  
  Longitude=c(80.2707,77.5946,78.4867)  
)
```

```
# Table
```

```
print(geo)
```

```
# Scatter Plot
```

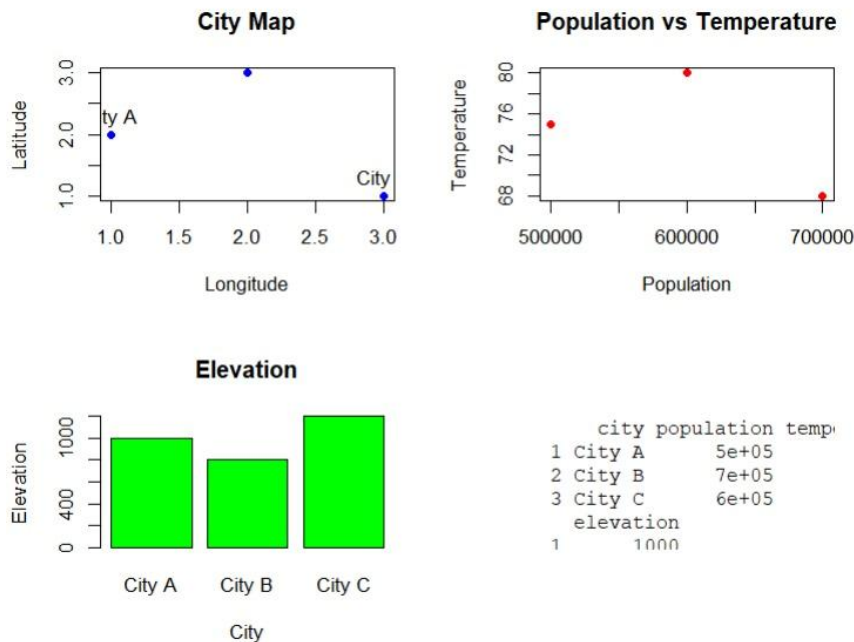
```
plot1 <- ggplot(geo,  
  aes(AvgTemperature,  
      Population,  
      label=City))+
```

```
geom_point(size=4,color="red")+
geom_text(vjust=-1)+
labs(title="Scatter Plot")
```

```
print(plot1)
```

```
# Map
```

```
leaflet(gео) %>%
addTiles() %>%
addMarkers(lng=~Longitude,
           lat=~Latitude,
           popup=~City)
```



SET 14 – Q1

Stacked Bar Chart (Question 1 Responses)

```
survey <- data.frame(
  Respondent=c(1,2,3),
  Question1=c("A","B","C"),
```

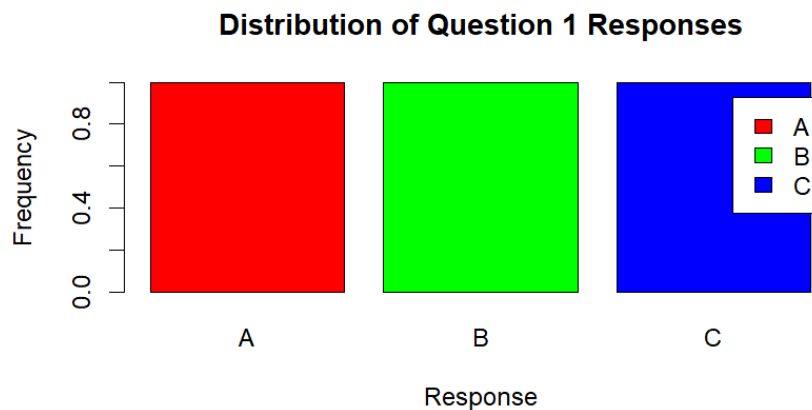
```

Question2=c("B","A","A"),
Question3=c("C","D","B")
)

response_count <- table(survey$Question1)

barplot(response_count,
        col=c("red","green","blue"),
        main="Distribution of Question 1 Responses",
        xlab="Response",
        ylab="Frequency",
        legend=rownames(response_count))

```



SET 14 – Q2

Generate a Radar Chart to visualize the overall pattern of responses.

Install package (Run only once)

```
install.packages("fmsb")
```

```
library(fmsb)
```

Dataset

```
survey <- data.frame(
```

```
  Respondent = c(1,2,3),
```

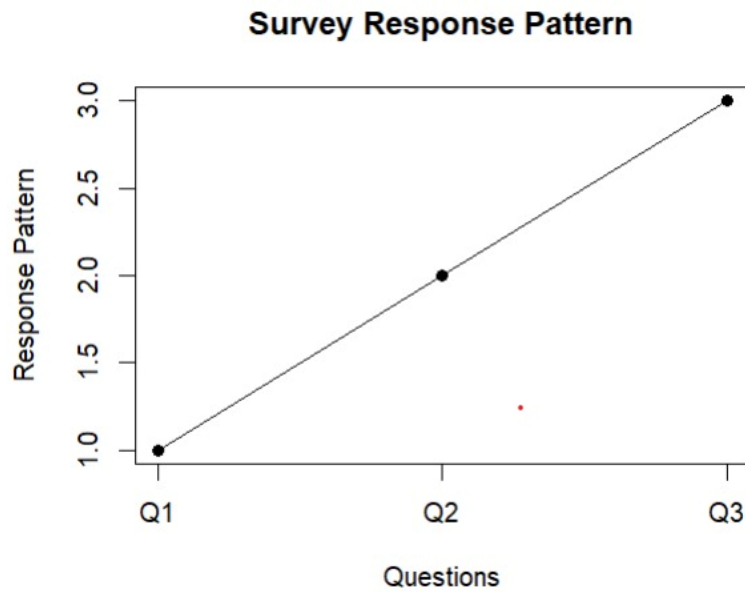
```
  Q1 = c(1,2,3),
```

```
  Q2 = c(2,1,1),
```

```

    Q3 = c(3,4,2)
)
# Radar Chart Data
radar_data <- rbind(
  c(4,4,4), # Maximum values
  c(0,0,0), # Minimum values
  survey[1,2:4],
  survey[2,2:4],
  survey[3,2:4]
)
colnames(radar_data) <- c("Q1","Q2","Q3")
radarchart(
  radar_data,
  axistype=1,
  pcol=c("red","blue","green"),
  pfc=c(rgb(1,0,0,0.3),
        rgb(0,0,1,0.3),
        rgb(0,1,0,0.3)),
  plwd=2,
  title="Radar Chart of Survey Responses"
)
legend(
  "topright",
  legend=c("Respondent 1","Respondent 2","Respondent 3"),
  col=c("red","blue","green"),
  lty=1
)

```



SET 14 – Q3

Build a table to show survey response data.

```
survey <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
```

```
cat("Survey Results\n\n")
```

```
print(survey)
```

```
> print(survey)
  Respondent Q1 Q2 Q3
1          1  A  B  C
2          2  B  A  D
3          3  C  A  B
```


SET 14 – Q4

Dashboard (Equivalent in R)

```
library(ggplot2)
```

```
library(fmsb)
```

```
survey <- data.frame(  
  Respondent = c(1,2,3),  
  Q1 = c("A","B","C"),  
  Q2 = c("B","A","A"),  
  Q3 = c("C","D","B")  
)
```

```
# Table
```

```
print(survey)
```

```
# Bar Chart
```

```
ggplot(survey,aes(Q1,fill=Q1))+  
geom_bar()+  
labs(title="Question 1 Responses",  
      x="Responses",  
      y="Count")
```

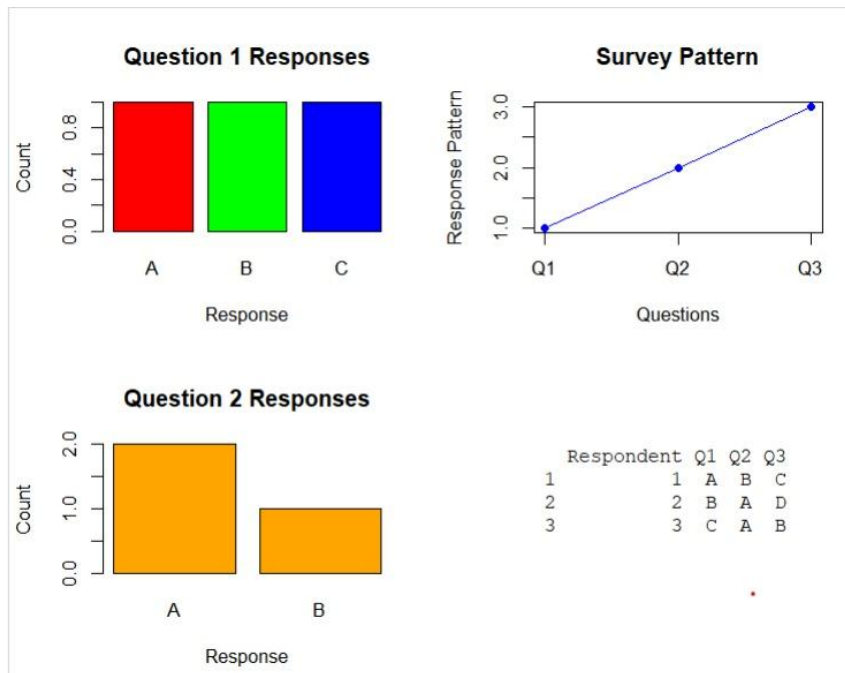
```
# Radar Chart
```

```
radar <- rbind(  
  c(4,4,4),  
  c(0,0,0),  
  c(1,2,3),  
  c(2,1,4),  
  c(3,1,2)  
)
```

```
colnames(radar)=c("Q1","Q2","Q3")
```

```
radarchart(radar)
```

```
cat("Dashboard Components Displayed Successfully\n")
```



SET 15 – Q1

Create a Grouped Bar Chart

```
install.packages("reshape2")
```

```
install.packages("ggplot2")
```

```
library(reshape2)
```

```
library(ggplot2)
```

```
sales <- data.frame(  
  Product=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100)
```

)

```
sales_long <- melt(sales,id.vars="Product")
```

```
ggplot(  
  sales_long,  
  aes(Product,value,fill=variable)  
)+  
  geom_bar(  
    stat="identity",  
    position="dodge"  
  )+  
  labs(  
    title="Quarterly Product Sales",  
    x="Product",  
    y="Sales"  
  )
```



SET 15 – Q2

Generate a Stacked Area Chart

```
library(reshape2)
```

```
library(ggplot2)
```

```
sales <- data.frame(  
  Product=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100)  
)
```

```
sales_long <- melt(sales,id.vars="Product")
```

```
ggplot(  
  sales_long,  
  aes(  
    x=variable,  
    y=value,  
    group=Product,  
    fill=Product  
  )  
) +  
  geom_area() +  
  labs(  
    title="Overall Sales Trend",  
    x="Month",  
    y="Sales"  
  )
```



SET 15 – Q3

Build a table to show the monthly sales data for each product.

Dataset

```
sales <- data.frame(  
  ProductID = c(1,2,3),  
  ProductName = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)
```

Display Table

```
cat("Monthly Product Sales Data\n\n")  
print(sales)
```

```
> print(sales)
  Product Jan  Feb  Mar
1 Product A 2000 2200 2400
2 Product B 1500 1800 1600
3 Product C 1200 1400 1100
```

SET 15 – Q4

Develop a dashboard combining the grouped bar chart, stacked area chart, and table.

Install packages (Run only once)

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
library(ggplot2)
```

```
library(reshape2)
```

Dataset

```
sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
```

Display Table

```
print(sales)
```

Convert data to long format

```
sales_long <- melt(sales, id.vars="Product")
```

Grouped Bar Chart

```
ggplot(sales_long,
  aes(Product, value, fill=variable)) +
  geom_bar(stat="identity", position="dodge") +
```

```
labs(title="Grouped Bar Chart",
      x="Product",
      y="Sales")
```

Stacked Area Chart

```
ggplot(sales_long,
      aes(variable, value,
           group=Product,
           fill=Product)) +
geom_area() +
labs(title="Stacked Area Chart",
      x="Month",
      y="Sales")
```

cat("Dashboard Components Displayed Successfully")

Note: Tableau dashboards cannot be created directly in R. This code displays the table and charts separately.



	Product	Jan	Feb	M
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

SET 16 – Q1

Create a bar chart to visualize the distribution of customer ages.

Install package (Run only once)

```
install.packages("ggplot2")
```

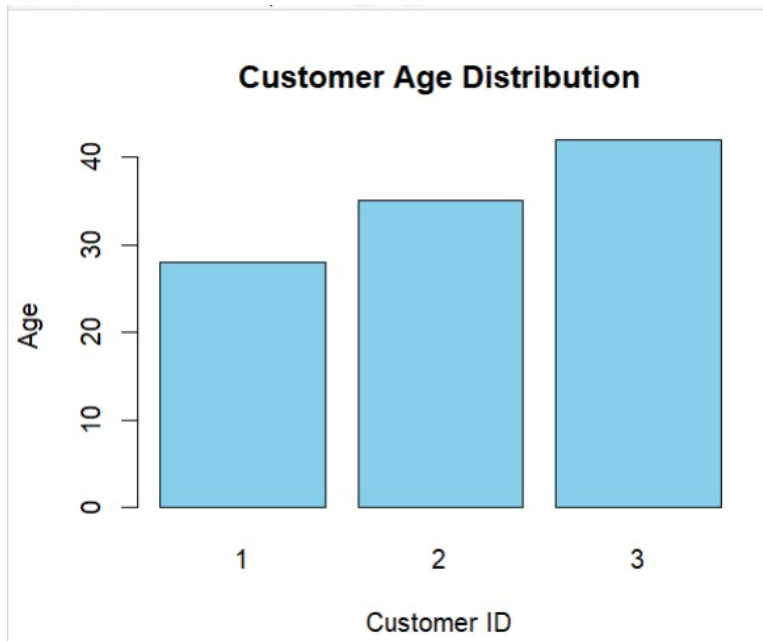
```
library(ggplot2)
```

Dataset

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

Bar Chart

```
ggplot(customer,  
  aes(x=factor(CustomerID),  
      y=Age,  
      fill=Gender)) +  
  geom_bar(stat="identity") +  
  labs(title="Customer Age Distribution",  
       x="Customer ID",  
       y="Age")
```

SET 16 – Q2

Generate a pie chart to represent the distribution of customers by gender.

Dataset

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

Count Gender

```
gender_count <- table(customer$Gender)
```

Pie Chart

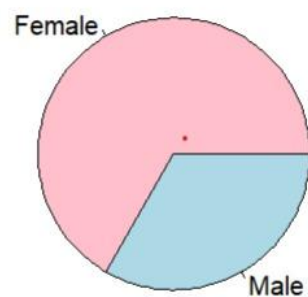
```
pie(  
  gender_count,  
  col=c("pink","lightblue"),  
  main="Customer Gender Distribution"  
)
```

```

legend(
  "topright",
  legend=names(gender_count),
  fill=c("pink","lightblue")
)

```

Gender Distribution



SET 16 – Q3

Build a table to show the demographic information for each customer.

Dataset

```

customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

```

```

cat("Customer Demographics\n\n")

```

```

print(customer)

```

	Customer	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

SET 16 – Q4

Develop a dashboard combining the bar chart, pie chart, and table.

Install packages (Run only once)

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

Dataset

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
```

Display Table

```
print(customer)
```

Bar Chart

```
ggplot(customer,
  aes(x=factor(CustomerID),
    y=Age,
    fill=Gender)) +
  geom_bar(stat="identity") +
  labs(title="Customer Age Distribution",
    x="Customer ID",
    y="Age")
```

```
# Pie Chart
```

```
gender_count <- table(customer$Gender)
```

```
pie(gender_count,  
    col=c("pink","lightblue"),  
    main="Gender Distribution")
```

```
cat("Dashboard Components Displayed Successfully")
```



SET 17 – Q1

Create a line chart to visualize the performance trend of employees over time.

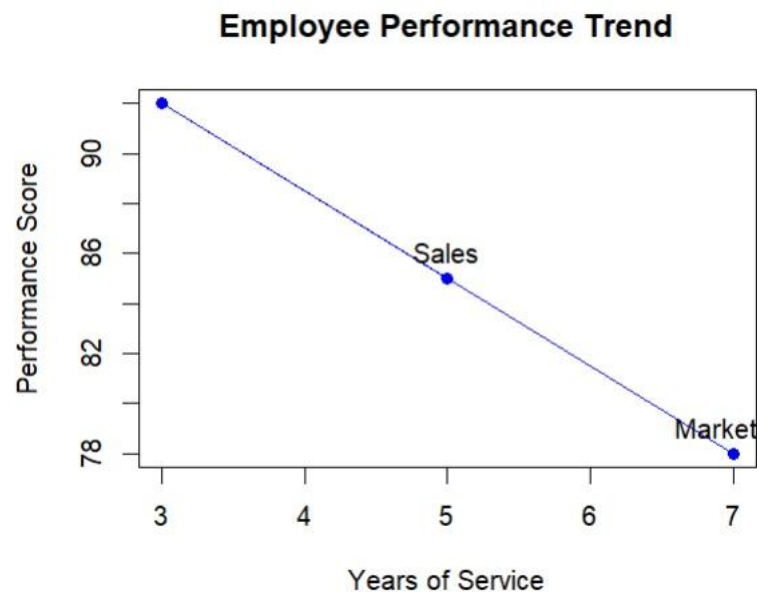
```
# Install package (Run only once)
```

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Dataset
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
```

```
# Line Chart
ggplot(employee,
  aes(x=YearsService,
      y=PerformanceScore,
      group=1)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(size=4, color="red") +
  labs(title="Employee Performance Trend",
       x="Years of Service",
       y="Performance Score")
```



SET 17 – Q2

Generate a bar chart showing the distribution of employees across different departments.

Install package (Run only once)

```
install.packages("ggplot2")
```

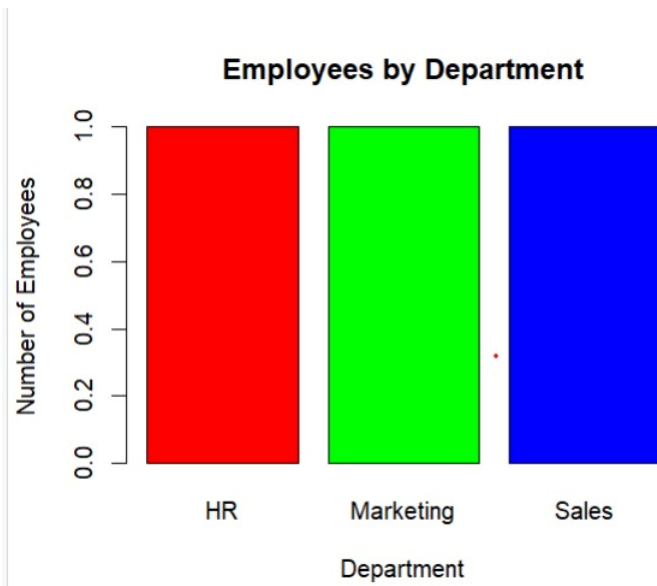
```
library(ggplot2)
```

Dataset

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)
```

Bar Chart

```
ggplot(employee,  
  aes(x=Department,  
    fill=Department)) +  
  geom_bar() +  
  labs(title="Employees by Department",  
    x="Department",  
    y="Number of Employees")
```



SET 17 – Q3

Build a table to display the performance data for each employee.

Dataset

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
```

```
cat("Employee Performance Data\n\n")
```

```
print(employee)
```

```
print(employee)
EmployeeID Department YearsService PerformanceScore
1          1      Sales           5             85
2          2        HR           3             92
3          3  Marketing           7             78
```

SET 17 – Q4

Develop a dashboard combining the line chart, bar chart, and table.

```
# Install package (Run only once)
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Dataset
```

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
```

```
# Display Table
```

```
print(employee)
```

```
# Line Chart
```

```
ggplot(employee,
  aes(x=YearsService,
      y=PerformanceScore,
      group=1)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(size=4, color="red") +
  labs(title="Performance Trend",
      x="Years of Service",
      y="Performance Score")
```

```
# Bar Chart
```

```
ggplot(employee,
  aes(x=Department,
      fill=Department)) +
  geom_bar() +
```



```
labs(title="Employees by Department",
      x="Department",
      y="Count")
```

```
cat("Dashboard Components Displayed Successfully")
```

